Objective:

This project analyses how customer engagement is distributed across departments, categories, and product types, and evaluates whether pricing aligns with customer satisfaction using engagement-weighted metrics.

Key Insights:

Higher-priced categories show relatively lower engagement and only average customer ratings, indicating potential value-perception challenges.

The **top three categories** account for **53% of total engagement**, indicating moderate structural concentration within the product portfolio**.** More than half of customer interaction is driven by a limited number of categories.

More than 60% Premium product pricing does not consistently result in higher customer ratings.

Several high-engagement and premium products have below-average ratings, indicating potential low performance in terms of quality and pricing.

Product performance varies significantly by category and price segment.

Business Implications:

The discounting strategy should be rethought.

Engagement concentration creates portfolio risk.

Quality improvement should be prioritised for high-visibility products; the premium products are performing below average.

The company should spread its focus across more categories instead of depending on just a few.

Business Problem Definition:

The company must concentrate on whether its pricing and discount strategies are effectively driving customer satisfaction and engagement.

Strategic risks are:

1. Engagement Concentration
   If most engagement comes from only a few categories, revenue could become unstable.

2. Pricing-rating misalignment
   If higher prices don't lead to higher customer satisfaction, customers may not see the product as worth the cost. 60% of the premium price segment has low performance, with over 200+ such products.

3. Ineffective discount strategy
   If discounts do not significantly increase engagement, promotional spending may not be used effectively. Even when the final price is lower, engagement still remains weak.

Data Overview:

The analysis was conducted using a structured product-level dataset containing pricing, rating, discount, and engagement metrics.

Key Variables:

Actual Price

Final price (Discounted price)

Discount Percentage

Rating

Rating Count

Product hierarchy (Department-Category-Product Type)

Key KPI frameworks:

Total engagement

Engagement share

Weighted average rating

Engagement-weighted revenue proxy

Price-rating position

Limitations:

Actual sales revenue was not available. So the rating count was used as an engagement and demand proxy.

Key Findings:

Pricing Distribution

Pricing variables were strongly right-skewed, indicating the presence of a limited number of high-priced premium products alongside a majority of lower-priced items.

Engagement Patterns

Engagement distribution is concentrated at the category level. The top three categories contribute 53.7% of total engagement, demonstrating structural concentration within the portfolio.

Correlation patterns

Weak correlation between:

1. Discount percentage and engagement (Rating count)
2. Discounted price and engagement (Rating count)

Limitations

- Rating-count was used as a proxy for demand due to the lack of sales data.
- Right-skewed distributions may influence aggregate metrics.
- Weak statistical correlations indicate that additional variables may be required for deeper behavioural modelling.

Recommendations:

Re-evaluate Discount Strategy
Given the weak relationship between discount levels and engagement, promotional spending should be reassessed to improve return on discount investment.

Enhance Data Collection
Incorporate actual sales and customer behaviour metrics for more accurate demand modelling.

Improve Quality of High-Visibility Products
Low-engagement, below-average-rated yet high-pricing electronics products should be prioritised for quality improvement to protect brand perception and improve engagement levels.